Rice - Leaf blast

RAG record ADV-007 | canonical key: rice_leaf_blast

Record ID	ADV-007
Crop / crop key	Rice / rice
Condition / disease key	Leaf blast / leaf_blast
Main pathogen	Magnaporthe oryzae (syn. Pyricularia oryzae)
RAG aliases	rice leaf blast, rice blast, Magnaporthe oryzae, Pyricularia oryzae, spindle lesions

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- Leaf blast starts as small spots that expand into spindle-shaped lesions with a grey/ashy centre and darker brown margin.
- Many lesions can merge, causing large dry areas and a burnt appearance in severe cases.

Weather or field conditions that favour it

High humidity, frequent rain or dew, and excessive nitrogen favour blast development.

What to do now

- Avoid excess nitrogen and continue balanced crop nutrition.
- Scout frequently during humid weather. Treat early if disease is increasing on susceptible varieties.
- Use resistant varieties and clean seed where available.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Tricyclazole 75% WP	0.6 g/L water	TNAU weather-based rice disease manual lists this rate for blast during high-risk periods.	[S1]

Chemical / formulation	Rate from cited source	How to use this information	Source
Azoxystrobin	0.5 ml/L water	Alternative option listed in the same TNAU manual. Use a formulation currently registered for rice blast.	[S1]
Tricyclazole seed treatment	2 g/kg seed	ICAR Kharif Agro-Advisory 2025 lists this seed-treatment rate.	[S2]
Propiconazole 25% EC	500 ml/ha	ICAR Kharif Agro-Advisory 2025 lists this foliar rate for paddy blast.	[S2]

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

Prevention and integrated management

- Avoid excessive nitrogen.
- Use resistant/tolerant varieties.
- Use healthy seed and suitable spacing.
- Monitor nursery and main field so early leaf blast is not allowed to build up before heading.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] Tamil Nadu Agricultural University (TNAU). Important Diseases in Rice and Their Weather-Based Management.
https://www.agritech.tnau.ac.in/pdf/ifpri%20merfs/Final%20English%20Manual.pdf

[S2] Indian Council of Agricultural Research (ICAR). Kharif Agro-Advisory for Farmers 2025.
https://icar.gov.in/sites/default/files/Circulars/ICAR-En-Kharif-Agro-Advisories-for-Farmers-2025.pdf